

Summer Math Adventure

W1
D1

Week 1 · Day 1 | Place Value: Reading & Writing Big Numbers

Name: _____

Date: _____

PLACE VALUE

Remember: Every digit has a *place* AND a *value*.

In 4,728,013 — the digit **7** is in the **hundred-thousands** place. Its value is **700,000**.

Periods help us read numbers: | Billions | Millions | Thousands | Ones |

Part A – Name That Place

Write the **place** and **value** of the underlined digit.

1. 3,462,805

Place: _____

Value: _____

2. 27,914,600

Place: _____

Value: _____

3. 513,002,744

Place: _____

Value: _____

4. 8,004,371

Place: _____

Value: _____

Part B – Expanded Form

Write each number in expanded form. Example: $4,302 = 4,000 + 300 + 2$

5. 605,814 = _____

6. 3,070,009 = _____

7. Write the number for: $400,000 + 60,000 + 500 + 7 =$ _____

Part C – Comparing Numbers

Write $<$, $>$, or $=$ in each circle.

8. 3,841,002 3,841,020

9. 70,500,000 7,500,000

10. 999,999 1,000,000

11. 24,060,000 24,006,000

Part D – Rounding

12. Round 4,738,201 to the nearest **million**.

13. Round 583,647 to the nearest **ten-thousand**.

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D2

14. Round 29,450 to the nearest **thousand**.

15. Round 7,499,999 to the nearest **million**.

Name: _____

Date: _____

ROUNDING & ESTIMATION

Rounding Rule: Look at the digit **ONE** place to the **RIGHT** of where you're rounding.

- If it's 0–4 → round *down* (keep the digit the same, replace the rest with zeros)
- If it's 5–9 → round *up* (add 1 to the digit, replace the rest with zeros)

number and its

Part A – Round It!

Number	Nearest 10	Nearest 100	Nearest 1,000
4,763			
28,549			
305,281			
1,999,500			

Part B – Estimation in Action

*Estimate by rounding each number to its **leading digit**, then compute. Show both the estimate AND the exact answer.*

1. A school has 4,812 students and 3,276 moved in over summer. About how many students total?

Estimate:

Exact:

2. A factory makes 78,340 bolts per day. About how many bolts in 5 days?

Estimate:

Exact:

3. Phoenix gets about 8 inches of rain per year. Estimate rain over 12 years.

Estimate:

Exact:

Part C – Is the Estimate Reasonable?

Circle YES or NO and explain why.

4. Mia says $6,489 + 3,211 \approx 10,000$. YES / NO
Why?

5. Jake says $52,710 - 18,300 \approx 20,000$. YES / NO
Why?

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Part D – Number Patterns (Cumulative Review)

Week 1 • Day 3 | Addition & Subtraction: Large Numbers

Find the rule and fill in the blanks.

Name: _____

6. $3,000 \rightarrow 6,000 \rightarrow 12,000 \rightarrow \boxed{} \rightarrow \boxed{}$

Date: _____

Rule: _____

ADDITION & SUBTRACTION

Key Reminder: Always line up digits by *place value* before adding or subtracting.

When subtracting, if the top digit is smaller, you must **regroup (borrow)** from the next column.

Check: Addition answer – one addend = other addend. Subtraction answer + subtracted number = starting number.

Part A – Solve It (Show all work in the space)

1. $485,372 + 246,918 =$

2. $3,004,000 - 1,867,425 =$

★ Challenge – Peoria Population Puzzle

Peoria, AZ had a population of 190,985 in 2020. Mesa had 504,258. Chandler had 261,165.

A. Round each city's population to the nearest ten thousand.

Peoria:

3. $12,308 + 9,847 + 140,003 =$

4. $10,000,000 - 4,382,619 =$

Chandler:

B. About how many more people does Mesa have than Peoria and Chandler *combined*?

Answer:

Answer:

Work:

Part B – Error Hunt

Each problem below has a mistake. Find it, explain it, and write the correct answer.

5. A student solved: $500,000 - 238,471 = 272,439$

Correct answer:

What went wrong?

6. A student solved: $304,072 + 58,900 = 362,072$

Correct answer:

What went wrong?

7. The Grand Canyon had 4,532,677 visitors in one year and 5,969,811 the next. How many visitors total over both years? Round your answer to the nearest hundred-thousand.
Exact:

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D4

Rounded:
Week 1 Day 4 | Multiplication: Patterns & Properties

Name: _____

Date: _____

MULTIPLICATION FOUNDATIONS

Properties to remember:

- **Commutative:** $6 \times 9 = 9 \times 6$
- **Associative:** $(2 \times 3) \times 5 = 2 \times (3 \times 5)$
- **Distributive:** $7 \times 13 = 7 \times 10 + 7 \times 3 = 70 + 21 = 91$
- **Multiplying by 10, 100, 1,000:** Just add zeros! $47 \times 100 = 4,700$

Part A – Powers of 10 Patterns

1. $83 \times 10 = \boxed{}$

2. $7 \times 40 = \boxed{}$

$83 \times 100 = \boxed{}$

$7 \times 400 = \boxed{}$

$83 \times 1,000 = \boxed{}$

$70 \times 400 = \boxed{}$

Challenge – Missing Digits

Fill in the missing digits to make each equation true.

A. $4 _ 7 _ 32 + 18 _ 40 _ = 635,837$

B. $500 _ 0 _ 0 _ 0 + 3 _ 5 _ 6 _ 9 = 5,356,900$

Part B – Distributive Property

Break apart each problem and solve. Show all steps.

3. 6×47
 $= 6 \times (_ + _)$
 $= _ + _$
 $= $

4. 8×93
 $= 8 \times (_ + _)$
 $= _ + _$
 $= $

5. 5×108
 $= 5 \times (_ + _)$
 $= _ + _$
 $= $

6. 12×25
 $= 12 \times (_ + _)$
 $= _ + _$
 $= $

Part C – Multi-Digit Multiplication

7. $346 \times 7 =$

8. $524 \times 38 =$

9. $2,004 \times 6 =$

Part D – Word Problem (Cumulative)

10. Destiny's school uses 48 boxes of pencils each month. Each box has 144 pencils. The school year has 9 months. How many pencils does the school use in a full year? First estimate, then find the exact answer.

Estimate:

Exact:

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Week 1 · Day 5 | Division: Meaning, Remainders & Word Problems

Name: _____

Date: _____

DIVISION

Division = splitting into equal groups.

$372 \div 6$: "How many groups of 6 fit in 372?"

Remainder: the leftover amount that can't form a full group.

Check: (quotient $\times$ divisor) + remainder = dividend $\rightarrow (62 \times 6) + 0 = 372 \checkmark$

Let's find two different ways to use the distributive property to solve $372 \div 6$.

Part A – Division with Remainders

1. $97 \div 4 =$

2. $253 \div 7 =$

3. $8,145 \div 9 =$

Q: _____

Q: _____

Q: _____

R: _____

R: _____

R: _____

Part B – What Does the Remainder Mean?

For each problem, decide: Do you round UP, round DOWN, or USE THE REMAINDER?

4. 75 kids need to take the school van. Each van holds 8 kids. How many vans are needed?

Work: $75 \div 8 = \underline{\hspace{1cm}} \text{ R } \underline{\hspace{1cm}} \rightarrow$ Answer:

_____ vans (Why?

_____)

5. 130 cookies are packed equally into boxes of 12. How many *full* boxes can be packed?

Work: $130 \div 12 = \underline{\hspace{1cm}} \text{ R } \underline{\hspace{1cm}} \rightarrow$ Answer:

_____ full boxes (leftover cookies: $\underline{\hspace{1cm}}$)

6. 4 friends share \$47 equally. How much does each get, and how much is left over?

Each gets: \$ _____

Leftover: \$ _____

Part C – Long Division Practice

7. $4,368 \div 12 =$

8. $3,525 \div 15 =$

Week 1 Quiz – Number Sense & Operations

W1
QUIZ

Answer:

Topics: Place Value · Rounding · Addition · Subtraction · Multiplication · Division

Answer:

Name: _____

Date: _____

Score: _____ / 20

Part D – Cumulative Review Sprint (2 min)

Show all work. Each question is worth 1 point unless marked otherwise.

9. $403,002 + 298,998 =$

10. Round 7,654,321 to

11. $9 \times 800 =$

nearest million:

Section 1 – Place Value & Rounding (5 pts)

1. What is the value of the digit 6 in 46,803,210?

2. Write $5,000,000 + 30,000 + 400 + 9$ as a standard number.

★ Challenge – Division Puzzle

A number divided by 8 gives quotient 47 and remainder 3. What is the number?

3. Round 8,372,649 to the nearest hundred-thousand.

4. Which is greatest? Circle one:
2,047,000 2,470,000 2,407,000

A number divided by 6 gives a remainder of 4. The quotient is between 30 and 40. List all possible original numbers.

5. (2 pts) Write the number that is 100,000 MORE than 6,900,050.

Section 2 – Addition & Subtraction (5 pts)

6. $748,293 + 351,708 =$

7. $5,000,000 - 3,284,736 =$

8. (3 pts) A library has 2,403,811 books. They donate 385,200 and receive 740,000 new ones. How many books does the library have now?

Section 3 – Multiplication & Division (10 pts)

9. $463 \times 27 =$

10. $50 \times 700 =$

11. $3,276 \div 9 =$

12. $5,824 \div 16 =$

13. (2 pts) Use the distributive property to solve 14×35 . Show two steps.

14. (3 pts) 243 students are going on a field trip. Each bus holds 36 students. How many buses are needed? Will any bus be less than full — if so, how many seats are empty?

Buses needed:

Empty seats: